Arquitectura de Robots

SCARA Inverse Kinematics

Geometric Method

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Section 1.1

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SCARA Robot Structure

1 SCARA Robot (RRP) Inverse Kinematics

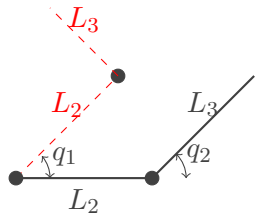
- 2 Revolute Joints (θ_1, θ_2)
- 1 Prismatic Joint (d)
- Planar XY motion + Vertical Z
- Cylindrical workspace

$$\theta_1 = q_1$$

$$\theta_2 = q_2$$

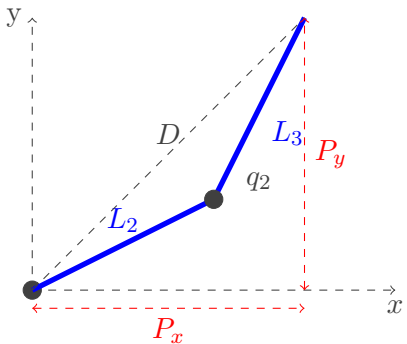
$$d_3 = q_3$$

Top view



Geometric Relationships

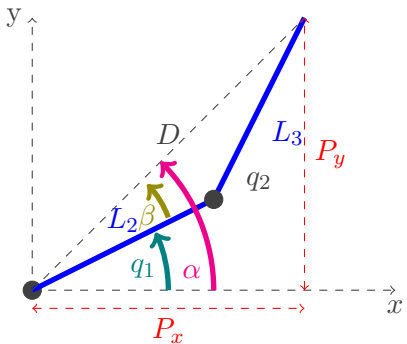
1 SCARA Robot (RRP) Inverse Kinematics



$$D = \sqrt{x^2 + y^2}$$

Step 1: Solve for q_1

1 SCARA Robot (RRP) Inverse Kinematics



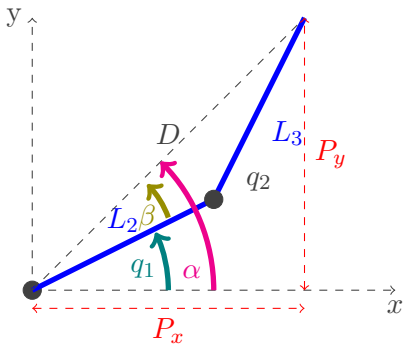
Solvin for α

$$\alpha = q_1 + \beta$$

Solvin for q_1

$$q_1 = \alpha - \beta$$

Step 2: Solve for α

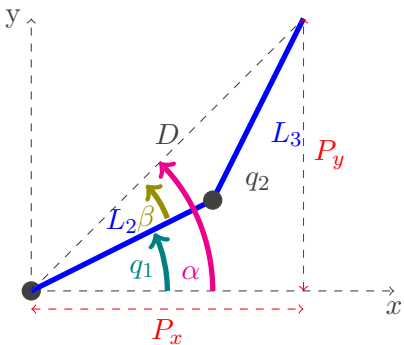


Utilizing the tangent, we are able to solve α

$$\tan(\alpha) = \frac{op}{ad}$$

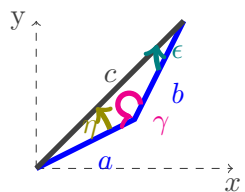
$$\alpha = tang^{-1}\left(\frac{P_y}{P_x}\right)$$

Step 3: Cosine law



to calculate β

Applying the cosine law

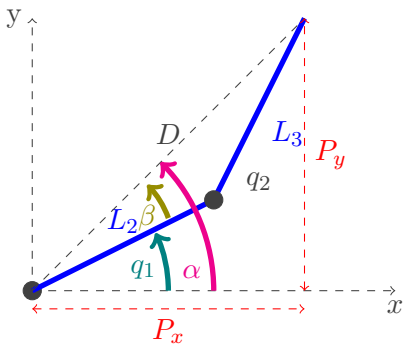


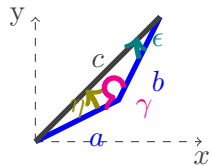
$$c^2 = a^2 + b^2 - 2ab \cdot cos(\gamma)$$

$$a^2 = c^2 + b^2 - 2cb \cdot cos(\epsilon)$$

$$b^2 = a^2 + c^2 - 2ac \cdot cos(\eta)$$

Step 4: Cosine law for β



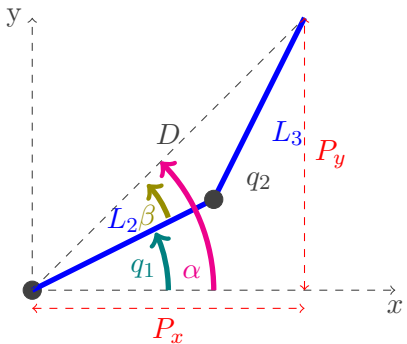


$$b^2 = a^2 + c^2 - 2ac \cdot cos(\eta)$$

substituting the equation into our robotic system model

$$L_3^2 = L_2^2 + D^2 - 2L_2D \cdot cos(\beta)$$

Step 5: Solve for β

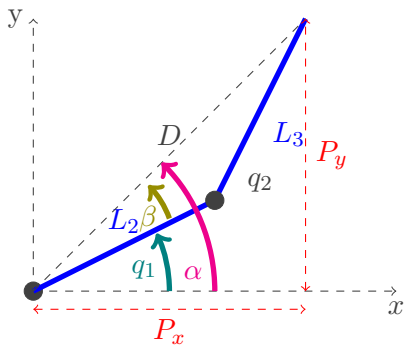


$$L_3^2 = L_2^2 + D^2 - 2L_2D \cdot cos(\beta)$$

solving for angle beta

$$\beta = cos^{-1}\left(\frac{L_3^2 - L_2^2 - D^2}{-2L_2D}\right)$$

Step 6: Substituting α and β



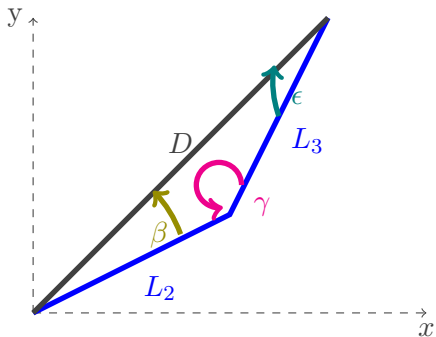
$$\alpha = tang^{-1}\left(\frac{P_y}{P_x}\right)$$

$$\beta = cos^{-1}\left(\frac{L_3^2 - L_2^2 - D^2}{-2L_2D}\right)$$

$$q_1 = \alpha - \beta$$

$$q_1 = tang^{-1}\left(\frac{P_y}{P_x}\right) - cos^{-1}\left(\frac{L_3^2 - L_2^2 - D^2}{-2L_2D}\right)$$

Step 1: Solve for q_2 (Law of Cosines)

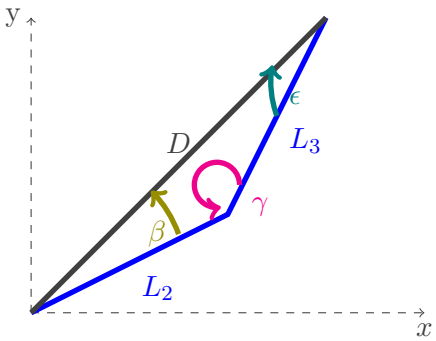


$$c^2 = a^2 + b^2 - 2ab \cdot cos(\gamma)$$

substituting the equation into our robotic system model

$$D^2 = L_2^2 + L_3^2 - 2L_2L_3 \cdot cos(\gamma)$$

Step 2: Solve for γ

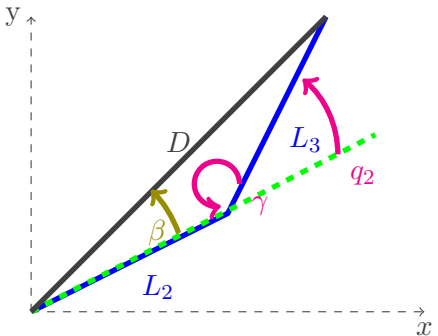


rearranging to isolate γ

$$D^2 = L_2^2 + L_3^2 - 2L_2L_3 \cdot cos(\gamma)$$

$$\gamma = \cos^{-1}\left(\frac{D^2 - L_2^- L_3^2}{-2L_2L_3}\right)$$

Step 3: Solve for q_2



if

$$180^\circ = \gamma + q_2$$

solving for q_2

$$q_2 = 180 - \gamma$$

$$q_2 = 180 - \cos^{-1}\left(\frac{D^2 - L_2^- L_3^2}{-2L_2L_3}\right)$$

Solve for d_3

Direct Relationship

$$d_3 = z$$

- Independent vertical coordinate
- Physical actuator constraints
- Consider initial offset if present

Numerical Example

1 SCARA Robot (RRP) Inverse Kinematics

Given:

- $L_1 = 20cm$, $L_2 = 10cm$
- Target position $(15, 20, 5)$

Solution:

Solve for D

$$D = \sqrt{15^2 + 20.0^2} = 20.5m$$

Solve for θ_2

$$\cos\phi_2 = \frac{20.5^2 - 20^2 - 10^2}{2 \cdot 20 \cdot 10} = 20.5$$

$$\theta_2 = 180 - \pm 75.52^\circ$$

Numerical Example

1 SCARA Robot (RRP) Inverse Kinematics

Given:

- $L_1 = 20cm,\ L_2 = 10cm$
- Target position $(15, 20, 5)$

Solution:

Solve for θ_1

$$\alpha = \arctan(20/15) = 53.13^\circ$$

$$\beta = \arccos\left(\frac{20^2 + 25^2 - 10^2}{2 \cdot 20 \cdot 25}\right) = 11.54^\circ$$

$$\theta_1 = 53.13^\circ \pm 11.54^\circ$$

$$d_3 = 5cm$$

Multiple Configurations

1 SCARA Robot (RRP) Inverse Kinematics

Elbow-up:

- $\theta_2 > 0$
- $\theta_1 = \alpha - \beta$



Elbow-down:

- $\theta_2 < 0$
- $\theta_1 = \alpha + \beta$



Important Considerations

1 SCARA Robot (RRP) Inverse Kinematics

- Singularities ($D = L_1 \pm L_2$)
- Joint limits
- Complex solutions = Outside workspace
- Configuration selection criteria

Conclusion

1 SCARA Robot (RRP) Inverse Kinematics

- Geometric method effective for planar robots
- Closed-form solution with two configurations
- Enables real-time control implementation
- Foundation for trajectory planning

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Q&A

Thank you for your attention

Rodriguez, J. Fundamentos De Robotica Antonio Barrientos.
Craig, J. J. (2006). Introduction to robotics. Pearson Educacion..
Spong, M. W., Hutchinson, S., & Vidyasagar, M. (2020). Robot modeling and control. John Wiley & Sons.